

TS Auto Research Agent Benchmark Report

1000 paper notes -> locked ETTh1 benchmark -> DLinear baseline 0.59827763 -> CalDLinear improves rmse by 0.00221968. PatchTST remains the stronger reference, so the claim stays bounded.

Current Claim

CalDLinear improves RMSE over DLinear by 0.00221968, but PatchTST remains stronger. This is a bounded candidate, not a SOTA claim.

Benchmark Results

baseline_anchor	DLinear	0.59827763	0.38131171	0	continue
strong_reference	PatchTST	0.58627319	0.37807289	0.01200444	continue
innovation_candidate	CalDLinear	0.59605795	0.38774657	0.00221968	continue

Literature Evidence

- APT: Affine Prototype-Timestamp For Time Series Forecasting Under Distribution Shift (AAAI_2026): Prototype or timestamp-conditioned affine structure motivates using known horizon context instead of only local instance statistics.
- Reversible Instance Normalization for Accurate Time-Series Forecasting Against Distribution Shift (ICLR_2022): Reversible normalization can help distribution shift, but its lookback-to-horizon statistic assumption must be tested.
- A Time Series is Worth 64 Words: Long-term Forecasting with Transformers (PatchTST) (ICLR_2023): Patch-based strong references prevent overstating a DLinear-only improvement.
- An Analysis of Linear Time Series Forecasting Models (ICML_2024): Recent linear-model analysis suggests constraints, normalization, and simple residual structure may matter more than adding heavy backbones.
- SIN: Selective and Interpretable Normalization for Long-Term Time Series Forecasting (ICML_2024): Selective normalization argues against one fixed statistic for every series and motivates gated lightweight adapters.

Next Action

Deepen 'CalDLinear' with ablations and more datasets before making a paper-level claim.